

Statistics

- Average ^{denotes} representative or typical values of whole set of obser. Since, typical values occupy central position, some obser. are larger & some others are smaller, averages are known as measures of central tendency.

→ Mean, median, mode

↓
A.M., G.M., H.M.

1. A.M. - Let a set of obser. denoted as x_i , simple A.M. ($\bar{x}$)

$$\bar{x} = \frac{1}{M} \sum_{i=1}^M x_i \text{ for the cardinality of set as } M.$$

Weighted A.M. ($\bar{x}$) = $\sum_{i=1}^M x_i f(x_i) x_i$, where f_i denotes freq. of x_i .

Again, if each group is $\bar{x}_i$ & with no. of obser. n_i , then

A.M. of composite group ($\bar{x}$) is defined as $\bar{x} \times N = \sum n_i \times \bar{x}_i$

where $N = \sum n_i$, So $\bar{x} = \frac{\sum n_i \times \bar{x}_i}{\sum n_i}$

Properties:

- Total of set of obser. = Prod. of their no. & A.M.

- Sum of deviations from A.M. is always 0.

$$\rightarrow \sum (x_i - \bar{x}) = 0$$

$$\rightarrow \sum f(i) (x_i - \bar{x}) = 0 \Rightarrow \sum f(x_i - \bar{x}) = 0$$

- If 2 var. x & y are relation ($y = Ax + B$), then

$$\bar{y} = A\bar{x} + B$$

If group of n_1 obser. has A.M. $\bar{x}_1$ & n_2 has A.M. $\bar{x}_2$

then A.M. of composite group $n_1 + n_2 = N$ (say)

is given by $N\bar{x} = n_1\bar{x}_1 + n_2\bar{x}_2$

- Sum of the sq. of deviations of a set of obser. has smallest value when deviation are from A.M.

$\sum (x_i - A)^2$ is min., where A is simplest mean.

2. $\sum f_i (x_i - A)^2$ is min. where A is weighted.

2. G.M. - When n obser. $x_1, \dots, x_n$. G.M. (simple) = $\sqrt[n]{x_1 \cdot x_2 \cdot \dots \cdot x_n}$

2. weighted G.M. = $(x_1^{f_1} \times x_2^{f_2} \times x_3^{f_3} \dots x_n^{f_n})^{1/n}$

where $n = \sum f_i$

Properties

- Product of group of n obser. to their n^{th} power

- Log of G.M. of set of obser. is an of their log

If $g_1, g_2, \dots$ is G.M. of several $n_1, n_2, \dots$ obser. then

G.M. composite group = $G = (g_1^{n_1} \times g_2^{n_2} \dots)^{1/N}$

where $\log G = \frac{1}{N} \sum$ Sum of all $(N = n_1 + n_2 + \dots)$

3. H.M. - Reciprocal of A.M. of their reciprocals.

G.M. & H.M. is defined iff no obs. $\neq 0$,

$$\text{H.M.} = \frac{M}{\frac{1}{x_1} + \frac{1}{x_2} + \dots + \frac{1}{x_m}} = \frac{M}{\sum \frac{1}{x_i}} \quad (\text{Simple})$$

$$\text{Weighted H.M.} = \frac{M}{\frac{f_1}{x_1} + \frac{f_2}{x_2} + \dots + \frac{f_m}{x_m}} = \frac{M}{\sum \frac{f_i}{x_i}}$$

It can be proved for any set, $A.M. \geq G.M. \geq H.M.$

i.e. for any 2 obser. $G.M.^2 = A.M. \times H.M.$

4. Median - Middlemost value when obser. are arranged in order of magnitude. No. of obser. $<$ median is same as no. $>$ than it. $\Rightarrow$ It divides into equal half.

From simple series,

- Data are arranged in order of magnitude. If no. of obser. is odd, Median $\rightarrow$ Middle most element
else Arithmetic mean of 2 middle

- C.F. corresponds to each distinct value is calc. as
 $F = \text{Value of var. corresponds to C.F. } \left(\frac{n+1}{2}\right) \rightarrow \text{Median}$
- From freq. dist. (group), Median is that value that corresponds C.F. $\frac{n}{2}$ as:

If class is in which C.F. $\frac{n}{2}$ lies $\rightarrow$ Median class

$$\therefore \text{Median} = A_1 + \frac{\frac{N}{2} - F}{f_n} \times C$$

$F = \text{C.F. below } A_1$

$f_n = \text{Freq. of median class}$

$C = \text{Width of median class}$

- L_1 & L_2 one formula

5. Mode - In a set of obser., mode is value that has max. freq. $\Rightarrow$ That value which has largest freq.

In group freq. dist. of all classes with equal width,

$$\text{Mode} = L_1 + \frac{D_1}{D_1 + D_2} \times C$$

where, $L_1 = \text{Lower boundary of modal class (class with } \uparrow \text{ freq.)}$

$D_1 = \text{Diff of largest freq. \& freq. of class preceed modal class}$

$D_2 = \text{Diff " just modal class}$

$C = \text{Class width}$

If $D_1 = f_0 - f_1$ & $D_2 = f_0 - f_1$ when f_0, f_{-1} & f_1

represent freq. of modal class, class preceed & following it

then $\text{Mode} = \frac{f_0 - f_{-1}}{2f_0 - f_{-1} - f_1} \times C$

* Relation:

Moderate skewness $\approx$

$$\text{Mean} - \text{Mode} \approx 3(\text{Mean} - 2\text{Median})$$

if $\bar{x}$ is the arithmetic mean of entity $x_1, x_2, \dots, x_n$
 then $x_i - \bar{x} = 0$ & $\sum f_i(x_i - \bar{x}) = 0$.

Q2. Find $\bar{x}$ from the table:

Commodity	Index No.	weight	$y = x - 100$	fy
A	92	12	-8	-96
B	128	7	28	-175
C	180	6	80	480
D	80	9	-20	180
		<u>34</u>		<u>379</u>

$$\text{So } \bar{y} = \frac{\sum fy}{N} = \frac{379}{34} = 11.11$$

$$C = 100$$

$$\bar{x} = C + \bar{y} = 100 + 11.11 = 111.11$$

Q3.

x :	55	65	75	85	95	105	115	
f :	8	10	16	14	10	5	2	$\sum f = 65$

$$y = \frac{x - 85.00}{10} : -3 \quad -2 \quad -1 \quad 0 \quad 1 \quad 2 \quad 3$$

$$fy: \text{ } \sum fy = -34$$

$$\bar{x} = 85.00 + 10\bar{y} \quad \bar{y} = \frac{\sum fy}{N}$$

$$= 85 + 10\left(\frac{-34}{65}\right) \approx 79.77$$

Q: Compute mean wt. of 65 from given table

Q4. Given, following dist. of monthly wage & no. of workers;
 Also calc. mean.

$C = 30.0$
 $d = 5$

Class Interval	freq.	Mid. value (x)	$y = \frac{x-30.0}{5}$	fy
12.5-17.5	2	15	-3	-6
17.5-22.5	22	20	-2	-44
22.5-27.5	10	25	-1	-10
	14	:	0	0
	3	:	1	3
	4	:	2	8
	6	:	3	
	1	:	4	
52.5-57.5	1	55	5	
	<u>63</u>			<u>-22</u>

$$y = \frac{x-30.00}{5}$$

$$\bar{y} = \frac{\sum fy}{N} = \frac{-22}{63}$$

$$\therefore \bar{X} = C + d\bar{y} = 30.00 + 5\left(\frac{-22}{63}\right) \approx 28.25$$

Q5. Find missing freq.

Class limits	f	Mid. value (x)	$y = \frac{(x-11.0)}{0.5}$	fy
9.3-9.7	2	9.5		
9.8-10.2	5	10.0	-3	-10
10.3-10.7	f_3	10.5	-2	-10
	f_4	11.	-1	f_3
	14	11.5	0	0
	6	12	1	14
	3	12.5	2	12
12.8-13.2	1	13	3	9
	<u>60</u> (Given)		<u>4</u>	<u>4</u>

$$\bar{X} = C + d\bar{y} \quad \& \quad \bar{X} = 11.09 \text{ (Given)}$$

$$\Rightarrow 11.09 = 11.0 + 0.5\left(\frac{23-f_3}{60}\right)$$

$$\therefore f_3 = 12.2$$

Now, f_3 must be whole no. $\Rightarrow f_3 = 12$

$$\therefore f_4 = 29 - 12 = 17 \quad (\because \sum f = 60)$$

Q6. G.M. of 10 obser. on a certain variable was calc. to be 16.2. It was later known that one obser. is recorded as 12.9 wrongly. If it was 21.9. Calc. correct G.M.

Soln: $\log \text{G.M.} = \text{A.M. of their log}$

$$\Rightarrow \log g = \frac{1}{N} \sum \log x_i$$

Now, product of 10 obser. including incorrect value = 16.2^{10} .

When incorrect is replaced, product would be

$$g^{10} = \frac{(16.2)^{10}}{12.9} \times 21.9$$

$$\therefore g \approx 17.08$$

Q7. Find median from freq. table

x	f	C.F.
0	7	7
1	44	51
2	33	86
3	16	102
4	9	111
5	4	115
6	1	116 = N

$\frac{N+1}{2} = 58.5$, which is > 51
but < 86 (As C.F.)
which is corres. to $x=2$

$\therefore \text{Median} = 2$

Q8. Find median class of table:

Class Boundary	f	C.F.
15-25	10	10
25-35	4	14
35-45	11	25
45-55	19	44
55-65	14	58
65-75	2	60
	2	62 = N

$\frac{N}{2} = 31$ lies b/w 25 & 44 & correct value

median must lie in 35 & 44.

Median class: 35-45

$$\text{So, } \frac{\text{Median} - 35}{45 - 35} = \frac{31 - 25}{44 - 25} \Rightarrow \text{Median} = 40.26$$

Q9. Table below gives press. of 250 men. The reading were made to nearest km. & central value is given.

B.P. (mm)

No. of men

Calc. mean & median.

60	4
65	5
70	31
75	39
80	114
85	30
90	28
95	2

$$C = 75, d = 5$$

Soln:

Class Boundary	Freq. (f)	$y = \frac{x-75}{5}$	fy	C.F.
60 57.5-62.5	5 4	-3	-12	4
65 62.5-67.5	5	-2	-10	9
70	31	-1	-31	40
75	39	0	0	79
80	114	1	114	193
85	30	2	60	223
90	28	3	78	248
95 90.5 92.5-97.5	2	4	8	250 = N
	250			

$$\text{Mean} = 75 + \frac{5 \times 204}{250} = 79.08 \text{ mm}$$

Found $\frac{N}{2} = 125$ exceeds C.F. 79 but < 193

$\therefore$ Median class 77.5-82.5

$$\therefore \text{Median} = 77.5 + \frac{12.5 - 79}{114} \times (82.5 - 77.5)$$

$$= 79.5 \text{ mm}$$

Q10. Find modal wt. of foll:

wt. 50-59.99 60-69.99 70-79.99 80-89.99 90-99.99

No. of 8 10 16 14 10 5 2

Soln: Class 70-79.99 has max. freq. 16 is modal class with lower boundary 69.995.

$\therefore$ Classes of equal width where, $N_1 = 69.995$

$$f_0 = 16, f_1 = 14, C = 10, f_{-1} = 10$$

$$\therefore \text{Mode} = 69.995 + \frac{16-10}{2 \times 16 - 10 - 14} \times 10 \approx 77.50 \text{ rs.}$$

Q11. The monthly profit are as follows:

<u>Profits per shop</u>	<u>No. of shops</u>
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0-100

12

100-200

18

200-300

27

300-400

20

400-500

17

500-600

6

Find modal class

2 mode.

Soln: Largest freq. 27 lies 200-300 $\rightarrow$ Modal class

$$\therefore \text{Mode} = 200 + \frac{9}{9+7} \times 100$$

$$= 256.25$$

$$L_1 = 200, D_1 = 27 - 18 = 9$$

$$D_2 = 27 - 20 = 7$$

$$C = 100$$

27/1/26

Moments

Given, n obs., $x_1, x_2, \dots, x_n$ & an arbitrary const. 'A' then first moment about A = $\frac{1}{n} \sum (x - A)$

2nd moment : $\frac{1}{n} \sum (x - A)^2$

q^{th} " : $\frac{1}{n} \sum (x - A)^q$

Sometimes, denoted as $\bar{M}_1, \bar{M}_2, \dots, \bar{M}_q$.

So, $\bar{M}_1 = \frac{1}{n} \sum (x - A) = \bar{x} - A$

• Moments about zero : 1st moment about 0 = $\frac{1}{n} \sum x = \bar{x}$
 q^{th} " : $\frac{1}{n} \sum x^q$

• Moments about mean (central):

1st moment = $\frac{1}{n} \sum (x - \bar{x}) = 0 = m_1$

2nd " = $\frac{1}{n} \sum (x - \bar{x})^2 = s^2 = m_2$

3rd moment $\xrightarrow{(m_3)}$ Skewness

4th moment $\xrightarrow{(m_4)}$ Kurtosis

For a freq. dist., r^{th} ^{moment} $A = \frac{1}{N} \sum f(x-A)^r$

So, r^{th} about moment $(0) = \frac{1}{N} \sum f(x)^r$

& r^{th} central moment $= \frac{1}{N} \sum f(x-\bar{x})^r$ $[N = \sum f, \bar{x} = \frac{1}{N} \sum fx]$

⊕ Relation b/w central & non-central:

If non-central ^{moments} $\bar{m}_1, \bar{m}_2, \dots$ about arbitrary origin A are known then central moments can be obtained by $\bar{m}_2 = \bar{m}_2 - \bar{m}_1^2$

$\bar{m}_3 = \bar{m}_3 - 3\bar{m}_2 \times \bar{m}_1 + 2\bar{m}_1^3$, If $\bar{m}_1$ & $\bar{m}_2$

are ^{moments} ~~known~~ about origin A then $\bar{x} = \bar{m}_1 + A$, $\sigma^2 = \bar{m}_2 - \bar{m}_1^2$

Q1. The first 2 moments about 5 of the variable ~~are~~ ² and 20.

Find mean & variance

Soln: $\bar{m}_1 = 2$ & $\bar{m}_2 = 20$, $A = 5$

$$\bar{x} = 2 + 5 \quad \& \quad \sigma^2 = 20 - 2^2 = 16$$

$$= 7$$

Q2. The mean of certain dist. is 5. 2nd & 3rd ~~are~~ moments are 20 & 140. Find 3rd moment about 10.

Soln: Given, $\sum \frac{(x-5)^2}{n} = 20$ & $\sum \frac{(x-5)^3}{n} = 140$

We need to obtain $\sum x - 10$

$$\sum \frac{(x-5)^3}{n} - 15 \sum \frac{(x-5)^2}{n} + 75 \sum \frac{(x-5)}{n} - 125$$

$$= 140 - 15 \times 20 + 75 \times 0 - 125$$

$$= -285$$

Q3. If $\bar{m}_2$ & m_2 are respec. 2nd moment about origin A & that of $\bar{A}$, show $\bar{m}_2 = m_2 + d^2$, $d = \bar{x} - A$

Soln: $\bar{m}_2 = \frac{\sum (x-A)^2}{n}$, $m_2 = \frac{\sum (x-\bar{x})^2}{n}$

So, $\frac{\sum (x-A)^2}{n} = \frac{\sum (x-\bar{x})^2}{n} + \frac{\sum (\bar{x}-A)^2}{n} + 2 \frac{\sum (\bar{x}-A) \times \sum (x-\bar{x})}{n}$

$\therefore \bar{m}_2 = m_2 + \frac{\sum (\bar{x}-A)^2}{n} + 2 \frac{(\bar{x}-A) \cdot 0}{n}$

$= m_2 + d^2$

Q4. If m_R & α_R central moments of std

$Z = \frac{x - \bar{x}}{\sigma}$, show $\alpha_R = \frac{m_R}{\sigma^R}$

Soln: $m_R = \frac{\sum (x_i - \bar{x})^R}{n}$ & $\alpha_R = \frac{\sum (z_i - \bar{z})^R}{n}$

& $z_i = \frac{x_i - \bar{x}}{\sigma}$

So, $\bar{z} = \frac{1}{n} \sum z_i = \frac{1}{n} \sum \frac{(x_i - \bar{x})}{\sigma} = \frac{1}{n\sigma} \sum (x_i - \bar{x}) = 0$

So, $\alpha_R = \frac{\sum (z_i - 0)^R}{n} = \frac{\sum z_i^R}{n} = \frac{1}{n} \sum \left(\frac{x_i - \bar{x}}{\sigma} \right)^R$

~~$\frac{1}{n}$~~ $= \frac{1}{n\sigma^R} \sum (x_i - \bar{x})^R = \frac{1}{\sigma^R} \frac{\sum (x_i - \bar{x})^R}{n}$

$= \frac{m_R}{\sigma^R}$ (Proved.)

Skewness: A freq. dist. is symm. if freq are symm dist. about mean when value of var. is equidistant from mean has equal freq.

ex: 10 15 20 25 30

f: 3 7 16 7 3

$$\therefore \bar{x} = 20$$

Eg. Eg. is symm. as mean, median & mode are eqd.

Asymmetrical:

ex: 5-8 9-12 13-16 17-20 21-24

f: 7 18 23 16 7

where, mean = median = Mode

Skewness is measure

$$\text{Skewness} = \frac{\text{Mean} - \text{Mode}}{\text{S.D.}} = \frac{3 \times (\text{Mean} - \text{Median})}{\text{S.D.}}$$

$$= \frac{Q_3 - Q_2 - (Q_2 - Q_1)}{(Q_3 - Q_2) + (Q_2 - Q_1)}$$

$$= \frac{n_{30}}{63}$$

Q1. Find suitable measure of skewness:

Annual sales: 0-20 20-50 50-100 100-250 250-500 500-

No. of falls: 20 50 69 30 25 90

Soln: $Q_1 = 39.95$, $Q_2 = 76.45$, $Q_3 = 204.75$

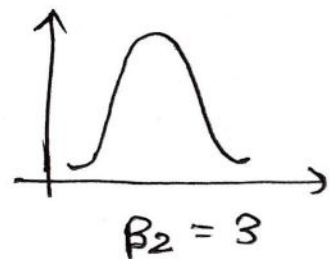
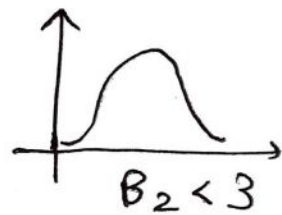
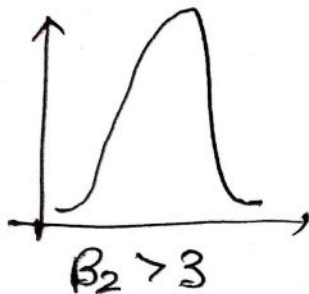
$$\text{Skewness} = \frac{Q_3 - Q_2 - (Q_2 - Q_1)}{(Q_3 - Q_2) + (Q_2 - Q_1)} = 0.55$$

• Kurtosis (Check spall.)

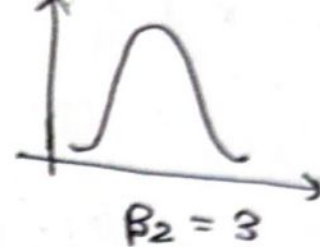
It is degree of peakness of curve. 2 ~~for~~ ^{dispersion} dists. has same avg. ^{skewness} but one may have high conc. of value of mode. This is called kurtosis. Measured by:

$$\frac{\mu_4}{\sigma^4} - 3 \quad (\text{Check})$$

formula



$$B_2 > 3$$



Correlation & Regression

Ex 1. R. Amt. of rainfall & yield of certain crop.

2. Height & wt. of a group of child.

3. Expen. of several families.

4. Age of husband & wife : 2 problems are involved

• Data ass. b/w x & y & we are interested to measure the strength of this association b/w variables. Linear or other eq. explains relationship b/w var. It is called co-variation.

• There may be one var. of an interest, other is auxiliary var.

Then, one may have interest using math. express. To estimate/predict the princ. var. This is called regression eq. &

prob. of making predictions is called regression.

Simple correlation & regression is linear.

* Data relating 2 var. is bivariate. Obser. on each individual are paired one for each - $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$,
say

The diagrammatic repres. is scatter diagram. It shows nature of 2 exp. of bivariate.
 $\rightarrow$ Eg +ve, -ve, 0 correlated.

• Correlation: It is used to denote
 Var. are called +ve correlation if $x \uparrow \& y \uparrow$.

~~It is used to denote~~

Given set of obs. : $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$ relating to var. x & y , $\text{cov}(x, y)$ is $\frac{1}{n} \sum (x - \bar{x})(y - \bar{y})$

$$\text{or } \text{cov}(x, y) = \frac{1}{n} \sum xy - \frac{\sum x}{n} \frac{\sum y}{n}$$

1. If $X = x - c, Y = y - c', c, c' \rightarrow \text{const.}$

$$\text{cov}(x, y) = \text{cov}(X, Y)$$

2. If $u = \frac{x - c}{d}, v = \frac{y - c'}{d'}, d, d' \rightarrow \text{const.}$

$$\text{cov}(x, y) = dd' \times \text{cov}(u, v), \text{ i.e. } x - \bar{x} = d(u - \bar{u}) = d(u - \bar{v})$$

Co-eff. (r): For a given n pairs of obs. of 2 obs. x & y ,

$$r = \frac{\text{cov}(x, y)}{\sigma_x \times \sigma_y} = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sqrt{\sum (x - \bar{x})^2 \times \sum (y - \bar{y})^2}}$$

$$= \frac{n \sum xy - \sum x - \sum y}{n \sum x^2 - (\sum x)^2 - n \sum y^2 - (\sum y)^2}$$

$$\text{So, } \text{var}(x) = d^2 \text{var}(u)$$

$$\text{var}(y) = d'^2 \text{var}(v)$$

Properties

If $u = \frac{x - \bar{x}}{s_x}$, $v = \frac{y - \bar{y}}{s_y}$

$r_{xy} = r_{yx}$, independent of choice

~~It is~~

- r lies b/w -1 to $1 \Rightarrow -1 \leq r \leq 1$
- r is ind. of unit, thus r is a pure no.

* Interpretation of r :

- Co-eff of correlation is degree of ass. b/w two var.
- Again, prop. of var. explained by regression = r^2 ~~prop.~~
- r also helps in est. the dependent var. when ind var. is known. Thus, for x , est. value of y from reg. of y on x

$$= \frac{y - y'}{r(s_y - b_x)}$$

If $r=0$, neither x & y can be est.

- r is also used in prob. of reliability in tests.
- Regression: It is used to denote est. or pred. of one var. for a specified value of other var. Est. is done by means of other eq., defined on bivariate data $\rightarrow$ Regression exp. Σ curve (geometrical rep.)

Linear regression when y is est. on x , $y' - \bar{y}$

$$b(y, x) = y - \bar{y} = x - \bar{x}$$

Est. of x by y , $\bar{x}' - \bar{x} = b(y' - \bar{y})$,

where b_{yx} = Reg. coeff of y on x .

Thus, $b_{yx} = \frac{\text{Cov}(x, y)}{\sigma_x^2} = \frac{r \sigma_y}{\sigma_x}$

$b_{xy} = \frac{\text{Cov}(x, y)}{\sigma_y^2} = \frac{r \sigma_x}{\sigma_y}$

* Prop. of linear eqⁿ:

- 2 linear eqⁿ given above where b_{yx} & b_{xy} are resp. of y on x & x on y .
- Prod. of 2 linear coeff of b_{yx} & $b_{xy} = r^2$
- r, b_{yx} & b_{xy} all have same sign if $r \neq 0$ then b_{yx} & b_{xy} are also 0.
- Reg. lines always intersect at $\bar{x}, \bar{y}$ & slopes of the lines b_{yx} & $\frac{1}{b_{xy}}$
- $\angle$ b/w 2 reg. lines depends on r . When $r = 0 \Rightarrow \perp$ li.
if $r = \pm 1 \Rightarrow$ Coincides
- 2 reg. eqⁿ are usually diff, when $r = \pm 1$, they are identical when $r = 0, y = \bar{y}$ & $x = \bar{x}$
& neither x or y can be est.

** Limitations:

- Bec. of closed correlation b/w x & linear eqⁿ, it is clear that former can serve as satisfactory measure of strength of 2 var. only when that relationship is linear. Hence, a low value of corr. coeff does not rule out poss. that var. are related in other manner. If any are non-linear, least sq. reg. lines will be parallel to co-ord. Hence, r will be very small.

Q1. Calc. coeff. of correlation (r):

x:	2.52	2.49	2.49	2.45	2.43	2.42	2.41	2.40
y:	730	710	710	890	970	1020	970	1040

Soln: $\therefore$ Coeff. of correlation is unaff. by origin & scale.

$$\text{Let } u = \frac{x - 2.45}{0.01}, v = \frac{y - 900}{100}$$

$\frac{x}{0.01}$	y	u	v	u^2	v^2	uv
2.52	730	7	-17	49	289	-119
2.49	710	4	-19	16	361	-76
2.49	710	4	-13	16	169	-52
2.45	890	0	-1	0	1	0
2.43	970	-2	7	4	49	-14
2.42	1020	-3	12	9	144	-36
2.41	970	-4	7	16	49	-28
2.40	1040	-5	14	25	196	-70
19.61	7100	1	-10	135	1258	-395

$$(\sigma_u)^2 = \frac{\sum u^2}{n} - \left(\frac{\sum u}{n} \right)^2 = \frac{135}{8} - \left(\frac{1}{8} \right)^2 = \frac{1039}{8}$$

$$(\sigma_y)^2 = \frac{1258}{8} - \left(\frac{1}{8} \right)^2$$

$$\text{Corr}(u, v) = \frac{\sum uv}{n \sigma_u \sigma_v}$$

$$= \frac{-395}{8} \times \frac{1}{\sqrt{\frac{1039}{8} \times \frac{1258}{8}}} = -0.64$$

Complete

Q2. n pairs are given (x_i, y_i) . Var. of x & var. of y are $\sigma_x^2, \sigma_y^2, \sigma_{xy}$. Show that $\sigma_{xy} = \frac{\sigma_x^2 - \sigma_y^2}{2\sigma_x\sigma_y}$

Soln:

~~Let~~ $u_i =$

$$u_i = x_i - y_i, \quad \bar{u} = \bar{x} - \bar{y}$$

$$u_i - \bar{u} = (x_i - y_i) - (\bar{x} - \bar{y}) = (x_i - \bar{x}) - (y_i - \bar{y})$$

$$\text{So, } \sigma_u^2 = \frac{\sum (u_i - \bar{u})^2}{n} = \frac{\sum (x_i - \bar{x})^2 - 2 \sum (x_i - \bar{x})(y_i - \bar{y}) + \sum (y_i - \bar{y})^2}{n}$$

$$= \frac{1}{n} \sum (x_i - \bar{x})^2 - 2 \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{n \sum (y_i - \bar{y})^2}$$

$$= \sigma_x^2 - 2 \sigma_x \sigma_y \times \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (y_i - \bar{y})^2}$$

$$= \sigma_x^2 + 2 \sigma_{xy} \sigma_y$$

Check & solve p2.

Q3 Find reg. of x on y from given data:

$$\sum x = 24, \sum y = 44, \sum xy = 306, \sum x^2 = 164, \sum y^2 = 574.$$

Find est. value of x when $y = 6$.

Soln: Eqⁿ: $x - \bar{x} = b_{xy} (y - \bar{y})$

$$b_{xy} = \text{Cov}(x, y) - \sigma_y^2$$

$$\text{Here, } \bar{x} = \frac{\sum x}{n} = \frac{24}{4} = 6, \quad \bar{y} = \frac{44}{4} = 11$$

$$\text{Cov}(x, y) = \frac{306}{4} - \frac{24}{4} \times \frac{44}{4} = 10.5$$

$$\sigma_y^2 = \frac{574}{4} - \left(\frac{44}{4}\right)^2 = 22.5$$

$$b_{xy} = \frac{10.5}{22.5} = 0.467$$

$$x - 6 = 0.467x(y - 11)$$

or

$$x = 0.467y + 0.$$

when $y = 6$, $x = 3.7$

Q4. Following results were obtained from age x & sys. B.P. of 10 women.

	x	y
Mean	53	142
Var.	130	165

Find app. reg. eq. & use it to find B.P. of age 45.

Soln: App. reg. eq. is reg. eq. of y on $x = y - \bar{y} = \frac{b_{xy}}{x} (x - \bar{x})$

$$b_{xy} = \frac{\text{Cov}(x, y)}{\sigma_x^2}$$

$$\text{Cov}(x, y) = \frac{130}{10}, \quad \text{Cov}(x, y) = 0.94$$

$$y = 0.94x + 32.18$$

when $x = 45$, $y = \underline{134.5}$ (Ans.)

Q5. Find eq. of line of reg. of x on y for following data:

x	1.0	1.5	2.0	2.5	3.0	3.5	4.0
y	53	57	63	72	82	87	84

Soln: Reg. of x on y : $x - \bar{x} = b_{yx}(y - \bar{y})$

We change origin for simplicity:

$$u = \frac{x - 2.5}{0.5}, \quad v = \frac{y - 72}{0.1}$$

$$\bar{x} = \frac{17.8}{7} \quad \bar{y} = \frac{49.8}{7}$$

$$\begin{aligned} \text{Cov}(u, v) &= \frac{\sum uv}{n} - \frac{\sum u}{n} \cdot \frac{\sum v}{n} \\ &= \frac{173}{7} - \frac{0}{7} \cdot \frac{8}{7} = \frac{172}{7} \end{aligned}$$

$$\sigma_v^2 = \frac{\sum v^2}{n} - \frac{(\sum v)^2}{n} = 7.96 \quad ?$$

$$\begin{aligned} \text{Cov}(x, y) &= 0.5 \times 0.1 \times \text{Cov}(u, v) \\ &= 0.5 \times 0.1 \times \frac{172}{7} \end{aligned}$$

$$\sigma_y^2 = (0.1)^2 \times \sigma_v^2 = (0.1)^2 \times 7.96$$

$$b_{xy} = \frac{\text{Cov}(x, y)}{\sigma_y^2} = \frac{0.8}{0.1 \times \frac{172}{7}} \times 7.96 \approx 0.76 \quad (\text{Check})$$

So, reg. line of x on y :

$$x - 2.5 = 0.76(y - 7.11)$$

$$\Rightarrow x = \frac{7.6}{2.90}$$

Q6. Let lines of reg. concerning 2 var. x & y , $y = 32 - x$ & $x = 13 - 0.25y$. Obtain mean & r .

Soln: $\therefore$ Reg lines int. at $(\bar{x}, \bar{y})$.

$$\text{Solving, } \bar{x} = 6.7, \bar{y} = 25.3$$

Now, $y = 32 - x$, so reg. of y on $x \Rightarrow b_{yx} = -1$

Again, $x = 13 - 0.25y$, $\therefore$ x on $y \Rightarrow b_{xy} = -0.25$

$$r^2 = b_{yx} \times b_{xy} = 0.25 \quad \therefore r = \pm 0.5$$

* (Note) Reg. coeff b_{yx} & b_{xy} both $-ve$, $\Rightarrow r = -0.5$ ✓

Q7. Regression of savings S on a family of income Y must be addressed as $S = a + \frac{1}{n} Y$, $a, n \rightarrow \text{const.}$ In a rand. sample of 100 families, var. of savings in one quarter & corr. is found to be 0.4. Find n .

Soln: S on Y : $S = a + \frac{1}{n} Y$ (Regg. eqn.)

Thus, regg. coeff of $S = \frac{1}{n}$

Now, $b_{xy} = r \frac{\sigma_x}{\sigma_y}$, $\sigma_x^2 = \frac{1}{4} \sigma_y^2$, $r = 0.4$

$\Rightarrow \frac{\sigma_x}{\sigma_y} = \frac{1}{2}$, Now, $b_{xy} = 0.4 \times \frac{1}{2} = 0.2$

$\Rightarrow 0.2 = \frac{1}{n} \therefore \underline{n = 5}$ (Ans.)

Q8. Given, $r = 0.8$, $\sum xy = 60$, $\sum x = 2.5$, $\sum x^2 = 19$.
Find no. of items

Soln: x & y denotes \$ den.

Lecture Notes: Sampling Theory and Regression

Course: Statistics

Date: February 12, 2026

Part 1 (0:00–10:00)

1. Regression (Questions First)

Question 1 (Problem 9): Find the number of items n

Given: $r = 0.8$, $\sum xy = 60$, $\sigma_x = 2.5$, $\sum x^2 = 90$

As stated in class (meaning of symbols): x and y denote deviations of X and Y from their means.

Equations Used

$$\text{Cov}(X, Y) = \frac{\sum xy}{n} = \frac{60}{n}$$

$$\sigma_X^2 = \frac{\sum x^2}{n} = \frac{90}{n}$$

$$r = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

Substitution written in class:

$$0.8 = \frac{60/n}{\sqrt{90/n} \times 2.5}$$

Solution

$$0.8 = \frac{60/n}{2.5\sqrt{90/n}}$$

$$0.8 \times 2.5 \times \sqrt{90/n} = 60$$

$$2\sqrt{90/n} = 60$$

$$\sqrt{90/n} = 30 \Rightarrow 90/n = 900 \Rightarrow n = 10$$

Answer: $n = 10$

Question 2: Regression coefficients when $u = x + y$, and correlation between x and $x + y$

Question (as dictated): If x and y are two correlated variables with same variance and correlation coefficient is r , find the regression coefficient of x on $(x + y)$, and that of $(x + y)$ on x ; hence find the correlation coefficient between x and $(x + y)$.

Let: $\text{Var}(x) = \text{Var}(y) = \sigma^2$

Also: $r = \frac{\text{Cov}(x, y)}{\sigma_x \sigma_y} \Rightarrow \text{Cov}(x, y) = r\sigma^2$ (since same variance)

Let: $u = x + y$

Equations Used

$$b_{xu} = \frac{\text{Cov}(x, u)}{\text{Var}(u)}, \quad b_{ux} = \frac{\text{Cov}(x, u)}{\text{Var}(x)}$$

$$\text{Var}(u) = \text{Var}(x + y) = \text{Var}(x) + \text{Var}(y) + 2 \text{Cov}(x, y)$$

$$\text{Cov}(x, u) = \text{Cov}(x, x + y) = \text{Var}(x) + \text{Cov}(x, y)$$

$$r_{xu} = \sqrt{b_{xu} b_{ux}}$$

Solution

$$\text{Var}(u) = \sigma^2 + \sigma^2 + 2(r\sigma^2) = 2\sigma^2(1 + r)$$

$$\text{Cov}(x, u) = \sigma^2 + r\sigma^2 = \sigma^2(1 + r)$$

$$b_{xu} = \frac{\sigma^2(1 + r)}{2\sigma^2(1 + r)} = \frac{1}{2}$$

$$b_{ux} = \frac{\sigma^2(1 + r)}{\sigma^2} = 1 + r$$

$$r_{xu} = \sqrt{\frac{1}{2}(1 + r)}$$

Question 3 (Problem 10A): Find the SD of y

Given: Karl Pearson's coefficient of correlation $r = 0.52$,
 $\text{Cov}(x, y) = 7.8$, and $\text{Var}(x) = 16$.

Required: Find σ_y .

Part 2 (10:00–20:00)

Question 3 Solution: Find the SD of y

Equations Used

$$r = \frac{\text{Cov}(x, y)}{\sigma_x \sigma_y}$$

$$\sigma_x = \sqrt{\text{Var}(x)}$$

Solution

$$\sigma_x = \sqrt{16} = 4$$

$$0.52 = \frac{7.8}{4\sigma_y}$$

$$\sigma_y = \frac{7.8}{0.52 \times 4} = \frac{7.8}{2.08} = 3.75$$

Answer: $\sigma_y = 3.75$ (or $15/4$)

Question 4 (Problem 10B): Find the SD of $x + y$

Given: Two correlated random variables x and y have SD 3 and 4 respectively, and $r = 0$.

Required: Find σ_{x+y} .

Equations Used

If $r = 0$, then $\text{Cov}(x, y) = 0$.

$$\text{Var}(x + y) = \text{Var}(x) + \text{Var}(y)$$

Solution

$$\text{Var}(x) = 3^2 = 9, \quad \text{Var}(y) = 4^2 = 16$$

$$\text{Var}(x + y) = 9 + 16 = 25$$

$$\sigma_{x+y} = \sqrt{25} = 5$$

Answer: $\sigma_{x+y} = 5$

2. Sampling Theory Definition

Why Sampling is Required (as mentioned)

The teacher explained that we may be given random variables (PDF also), and the question is how to predict the actual value. Using a wireless-channel example: the channel changes with each time slot (example duration: 1 millisecond), so data rate/power control changes, making it challenging. Hence sampling is important for practical experiments/projects/papers.

Definition (as dictated)

Sampling: "It denotes the selection of a part of the aggregate statistical material with a view to obtain information about the whole."

Population / Universe: "This aggregate (or totality of statistical information) on a particular character of all members covered by an investigation is called Population or Universe."

Sample: "The selected part which is used to ascertain the characteristics of population is called Sample."

3. Types of Sampling (Definitions Only, as Dictated)

Simple Random Sampling

"It is the process of selection of a group of units in such a manner that every unit of the population has an equal chance of being included in the sample."

Purposive Sampling

"A sample which is selected on the basis of individual judgment of the sampler is called Purposive Sample."

Stratified Sampling

"A population is subdivided in several parts called strata, and then sub-sample is chosen from each of them; all the sub-samples are combined together giving the stratified sampling."

Part 3 (20:00–30:00)

Multistage Sampling

"Sampling procedure which is carried out in several stages is called Multi-stage sampling."

4. Important Sampling Distributions

4.1 Random Sampling (Sampling Distribution of Sample Mean)

If $\bar{x}$ represents the mean of a random sample of size n drawn from a population with mean μ and standard deviation σ , then the sampling distribution of $\bar{x}$ is approximately a normal distribution with mean μ and standard deviation equal to the standard error of $\bar{x}$.

4.2 Sampling Distribution of Sample Proportion

If p represents the proportion of defectives in a random sample of size n drawn from a lot with proportion of defectives P , then the sampling distribution of p is approximately a normal distribution with mean P and standard deviation equal to the standard error of p .

Standard Error (SE)

Formulas (as dictated)

For random samples of size n :

$$SE(\bar{x}) = \frac{\sigma}{\sqrt{n}}$$

$$SE(p) = \frac{\sqrt{PQ}}{\sqrt{n}} = \sqrt{\frac{PQ}{n}}$$

where P = population proportion, $Q = 1 - P$

Properties (Only Those Mentioned)

Property 1 (Finite Population, Without Replacement)

If the random sample is drawn without replacement from a finite population of size N , then the above formulae are modified by multiplication by:

$$\sqrt{\frac{N-n}{N-1}}$$

Property 2 (Infinite Population / With Replacement)

If $\bar{x}$ is the mean calculated from a random sample of size n and the SD of the population is σ , then:

$$SE(\bar{x}) = \frac{\sigma}{\sqrt{n}}$$

when either the population size is infinitely large or the sample is drawn with replacement.

Property 3 (Finite N , Without Replacement)

For N finite (population size), but sample drawn without replacement:

$$SE(\bar{x}) = \frac{\sigma}{\sqrt{n}} \times \sqrt{\frac{N-n}{N-1}}$$

Property 4 (Proportion Cases)

For proportion P :

$$SE(p) = \sqrt{\frac{P(1-P)}{n}}$$

And with finite population correction (sample without replacement):

$$SE(p) = \sqrt{\frac{P(1-P)}{n}} \times \sqrt{\frac{N-n}{N-1}}$$

Part 4 (30:00–40:00)

Test of Significance (For Large Samples)

Basic Definitions (as dictated)

1. Null Hypothesis (H_0)

"A definite statement about the population parameter such as mean, proportion, etcetera is usually called Null Hypothesis; it is usually the hypothesis of no difference."

2. Alternative Hypothesis (H_1)

"Any hypothesis which is complementary to the Null Hypothesis is called Alternative Hypothesis."

3. Level of Significance (α)

"The probability of rejecting the Null Hypothesis when it is true is called Level of Significance; it is denoted by α ."

Level of significance is usually taken as 5% or 1% (0.05 or 0.01).

4. Critical Region (Region of Rejection)

"A region in the sample space which amounts to rejection of H_0 is called Critical Region or Region of Rejection."

Errors in Sampling (as dictated)

Type I Error

"If H_0 is rejected when it is true, this is Type I error."

Type II Error

"If H_0 is accepted when it is false (that means H_1 is true), this is Type II error."

General Procedure for Testing of Hypothesis (Steps)

Step 1

Set up H_0 and H_1 .

Step 2

Choose the level of significance α (generally 0.05 or 0.01).

Step 3 (Test Statistic)

Compute the test statistic Z under H_0 :

$$Z = \frac{t - E(t)}{SE(t)}$$

where t is the sample statistic (sample mean, sample proportion, etc.).

Step 4 (Comparison Rule)

- If calculated $|Z|$ is less than z_α , **accept** H_0 .
- If $|Z|$ is greater than z_α , **reject** H_0 .

Here z_α is the significant value of Z at α level of significance.

Critical Values of Z (Table Dictated)

Level of Significance	1% (0.01)	5% (0.05)	10% (0.10)
Two-tailed test	2.58	1.96	1.645
One-tailed test (right or left)	2.33	1.645	1.28

Part 5 (40:00–End)

5. Numerical Question on Sampling (Solution as Dictated)

Question

A sample of 400 individuals is found to have a mean height of 67.47 inches. Can it be reasonably regarded as a sample from a large population with mean height 67.39 inches and standard deviation 1.30 inches?

Given (as written in class)

$$n = 400, \quad \bar{x} = 67.47, \quad \mu = 67.39, \quad \sigma = 1.30$$

Step 1: Hypotheses (Two-Tailed)

$$H_0 : \mu = 67.39$$

$$H_1 : \mu \neq 67.39 \quad (\text{Two-tailed test})$$

Step 2: Test Statistic

$$Z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$$

Substitution (as done in class):

$$Z = \frac{67.47 - 67.39}{1.30 / \sqrt{400}}$$

$$Z = \frac{0.08}{1.30 / 20}$$

$$Z = \frac{0.08}{0.065} = 1.231$$

Step 3: Conclusion (5% Level, Two-Tailed)

$$|Z| = 1.231 < 1.96 \quad (\text{at 5\% level of significance})$$

Decision: Accept H_0

Conclusion: Hence, the sample can be regarded as taken from the given population.

End of Lecture Notes